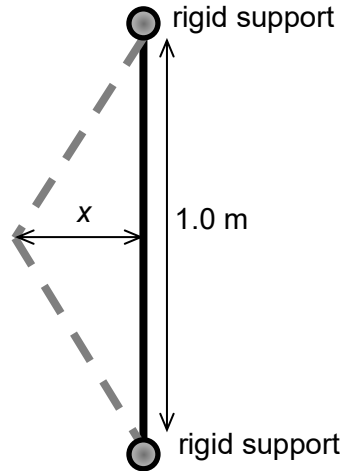


- 7 A student builds a launcher using an elastic cord of negligible mass and natural length of 1.0 m attached to two rigid supports.



Assuming that the elastic cord obeys Hooke's Law with a spring constant of 80 N m^{-1} and that the cord is pulled at its midpoint, what is the minimum draw length x needed such that a 200 g water balloon may be propelled with a speed of 10 m s^{-1} ?

- A** 0.50 m **B** 0.56 m **C** 0.75 m **D** 1.5 m

Ans: B

By COE,

Loss in EPE = Gain in KE

$$0.5ke^2 - 0 = 0.5mv^2 - 0$$

$$80(2\sqrt{0.50^2 + x^2} - 1.0)^2 = 0.200(10)^2$$

$$x = 0.56 \text{ m}$$

- 8 A simple idea for generating electricity from the tides allows water stored in a container at high